



This is not mandatory reading, but here's the code we'll run in the "Semi-Structured Data" videos. It may come in handy when you're doing the associated hands-on assignment.

---> see an example of a column with semi-structured (JSON) data

```
SELECT MENU_ITEM_NAME
      , MENU_ITEM_HEALTH_METRICS_OBJ
FROM tasty_bytes.RAW_POS.MENU;
```

```
DESCRIBE TABLE tasty_bytes.RAW_POS.MENU;
```

---> check out the data type for the menu_item_health_metrics_obj column - It's a VARIANT

*/

```
CREATE TABLE tasty_bytes.raw_pos.menu
(
  menu_id NUMBER(19,0),
  menu_type_id NUMBER(38,0),
  menu_type VARCHAR(16777216),
  truck_brand_name VARCHAR(16777216),
  menu_item_id NUMBER(38,0),
  menu_item_name VARCHAR(16777216),
  item_category VARCHAR(16777216),
  item_subcategory VARCHAR(16777216),
  cost_of_goods_usd NUMBER(38,4),
  sale_price_usd NUMBER(38,4),
  menu_item_health_metrics_obj VARIANT
);
*/
```

---> create the test_menu table with just a variant column in it, as a test

```
CREATE TABLE tasty_bytes.RAW_POS.TEST_MENU (cost_of_goods_variant)
AS SELECT cost_of_goods_usd::VARIANT
FROM tasty_bytes.RAW_POS.MENU;
```

---> notice that the column is of the VARIANT type

```
DESCRIBE TABLE tasty_bytes.RAW_POS.TEST_MENU;
```

---> but the typeof() function reveals the underlying data type

```
SELECT TYPEOF(cost_of_goods_variant) FROM tasty_bytes.raw_pos.test_menu;
```

---> Snowflake lets you perform operations based on the underlying data type

```
SELECT cost_of_goods_variant, cost_of_goods_variant*2.0 FROM  
tasty_bytes.raw_pos.test_menu;
```

```
DROP TABLE tasty_bytes.raw_pos.test_menu;
```

---> you can use the colon to pull out info from menu_item_health_metrics_obj

```
SELECT MENU_ITEM_HEALTH_METRICS_OBJ:menu_item_health_metrics FROM  
tasty_bytes.raw_pos.menu;
```

---> use typeof() to see the underlying type

```
SELECT TYPEOF(MENU_ITEM_HEALTH_METRICS_OBJ) FROM tasty_bytes.raw_pos.menu;
```

```
SELECT MENU_ITEM_HEALTH_METRICS_OBJ, MENU_ITEM_HEALTH_METRICS_OBJ['menu_item_id'] FROM  
tasty_bytes.raw_pos.menu;
```